Understanding

Hybrid Electric Vehicles

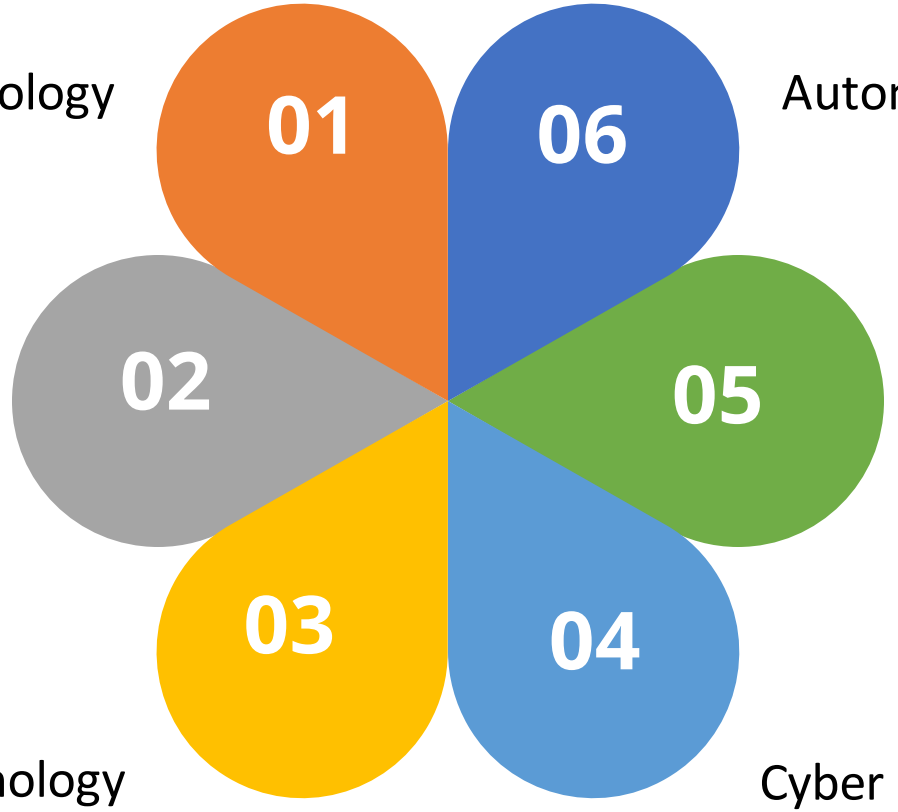


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Our Expertise



Electric Vehicle Technology

Automotive Electronics

Future Mobility Overview

IoT, Connectivity &

Advanced Computing

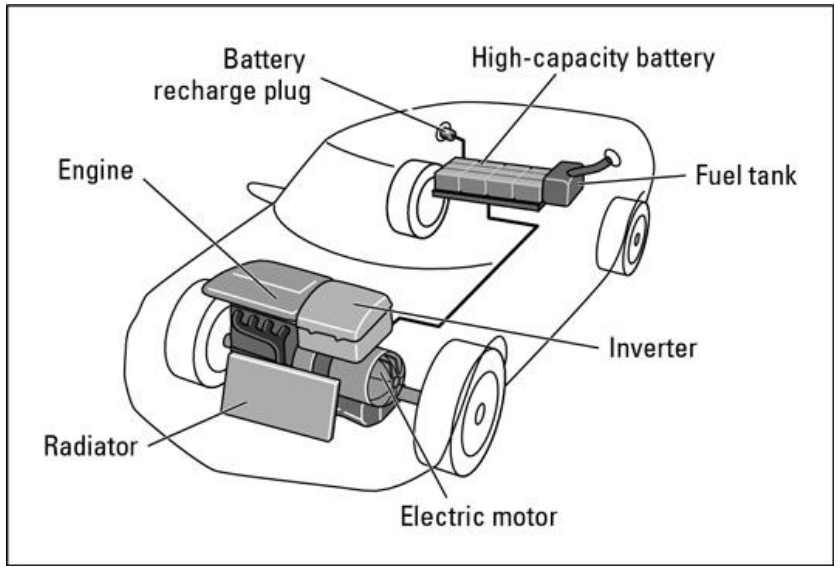
ADAS & Autonomous Technology

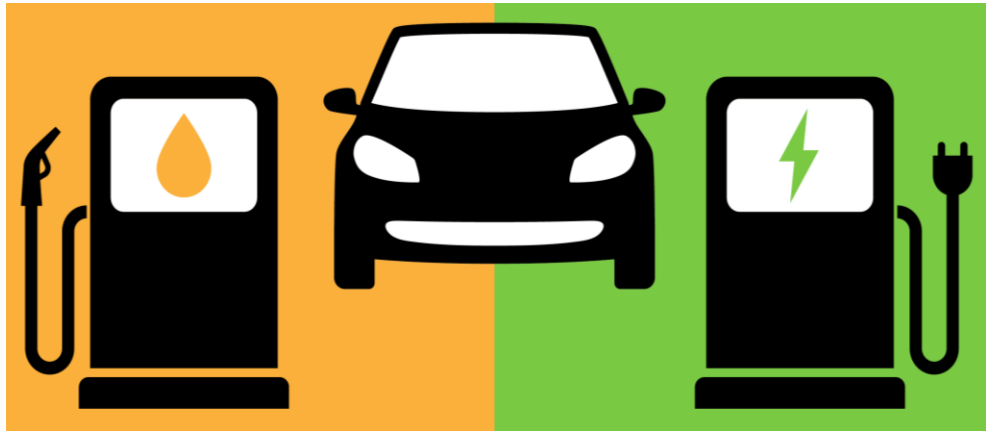
Cyber Security for Automotive

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What is a Hybrid Electric Vehicle (HEV) ?

- These are vehicle that using multiple energy source for propulsion.
- They uses an IC engine that converts heat into mechanical energy as primary source.





- Also, have a battery pack and motor to convert electrical energy into mechanical
- The motor can assist the engine or fully drive the vehicle depending on battery

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Why HEVs are needed ?

- IC engines are not efficient at low speeds, and produce peak torque at higher rpm.
- In these times, they emit more emissions and pollute air with CO, NOx, SO2, etc.

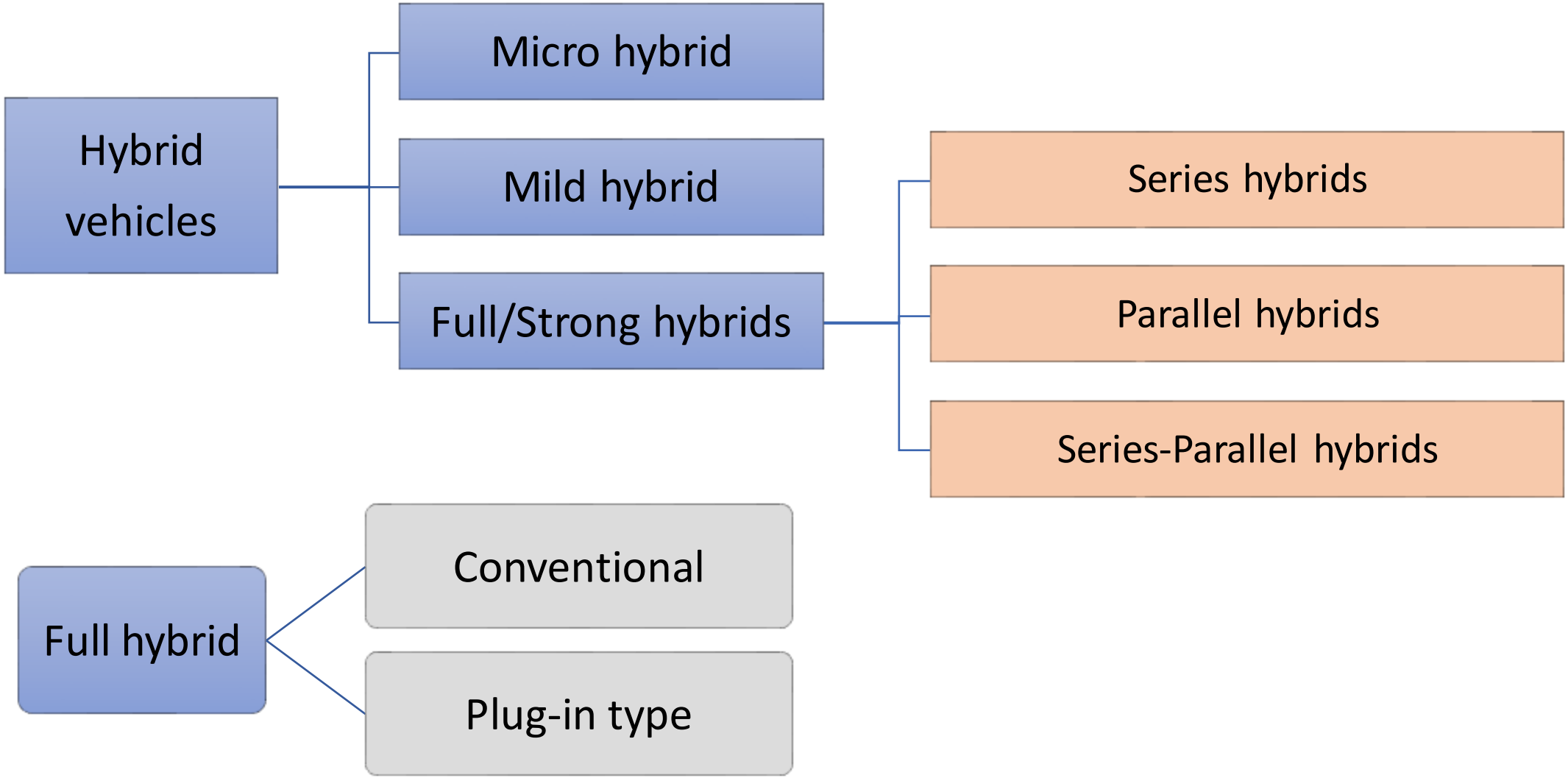




- The electric motors are highly efficient and produce high torque at low speeds/rpm.
- Thus, a combination of both will help to reduce emissions and increase fuel efficiency

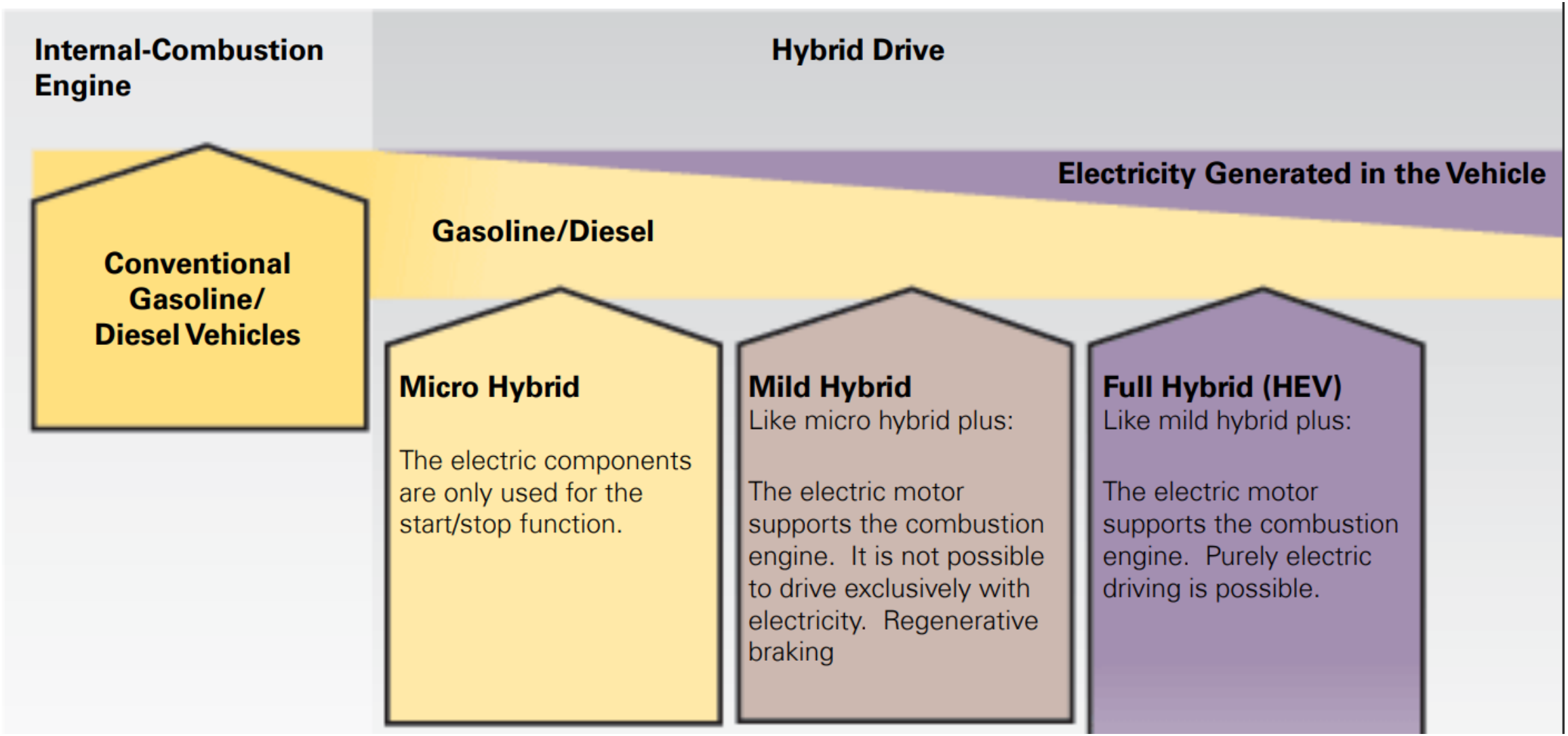
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Types of HEVs



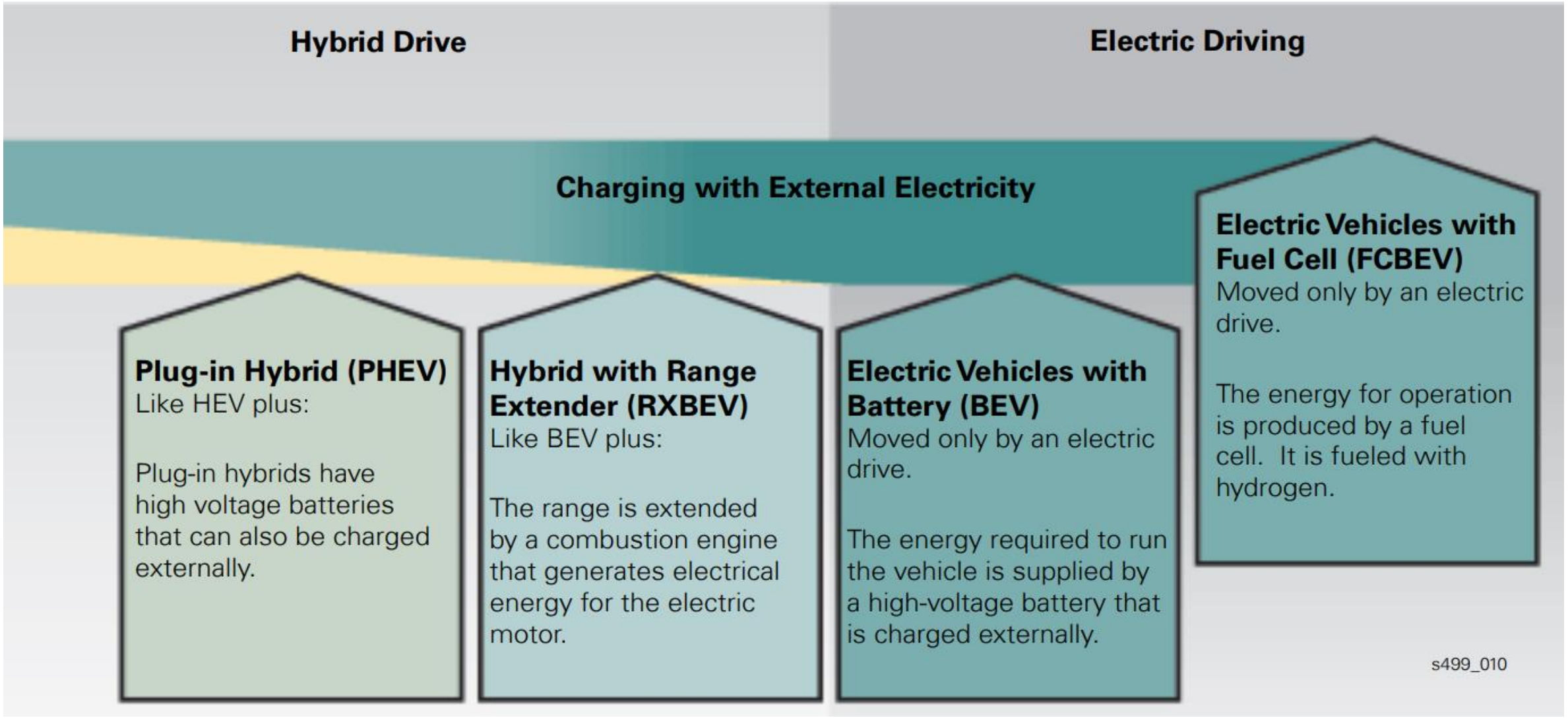
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Electrification of powertrain



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Going green with electrification



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What is Micro Hybrid ?

- Micro hybrid is just a start-stop system to save fuel and thus, not a real hybrid.
- It switches off the engine when idling for more than 10 sec. (if the gear in neutral)





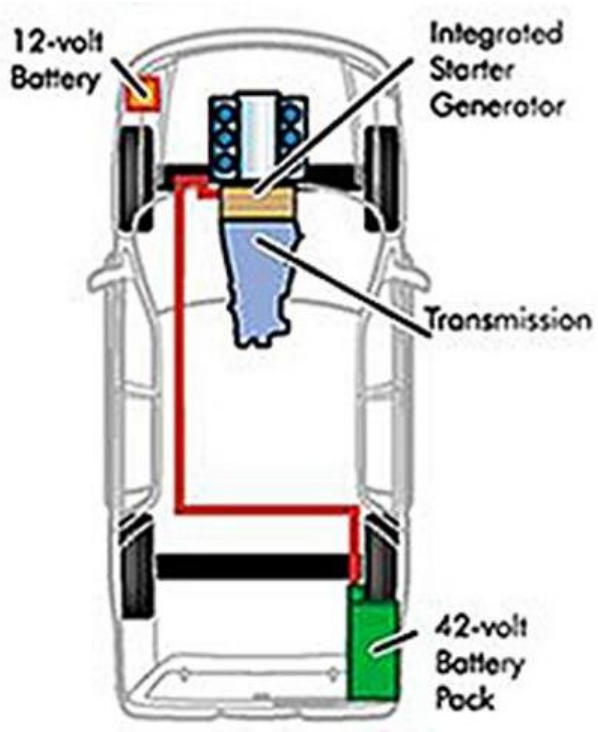




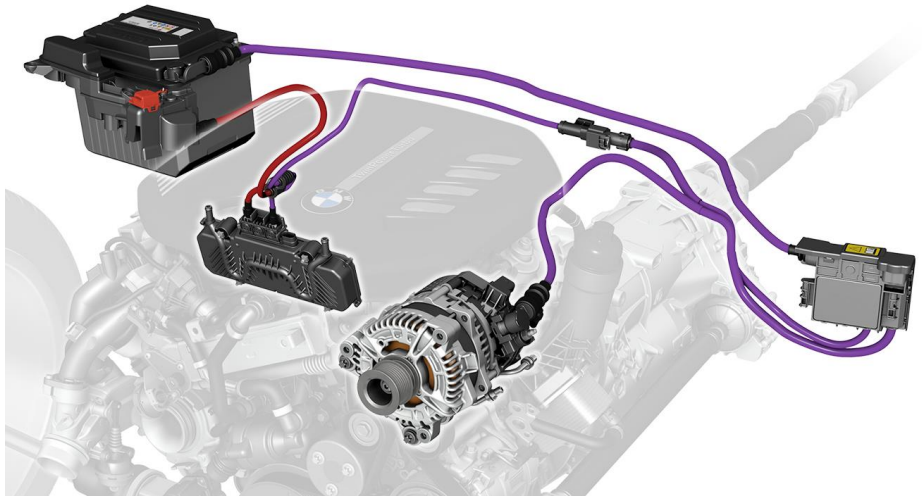
- Driver either press clutch pedal or accelerator to restart the engine for driving.
- Thus, it saves precious fuel lost during long idling situations like traffic blocks.

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What is Mild Hybrid ?



- This is the starting point of electric motor and additional battery storage in vehicles.
- The motor is to assist the engine while high load conditions, not for driving the vehicle

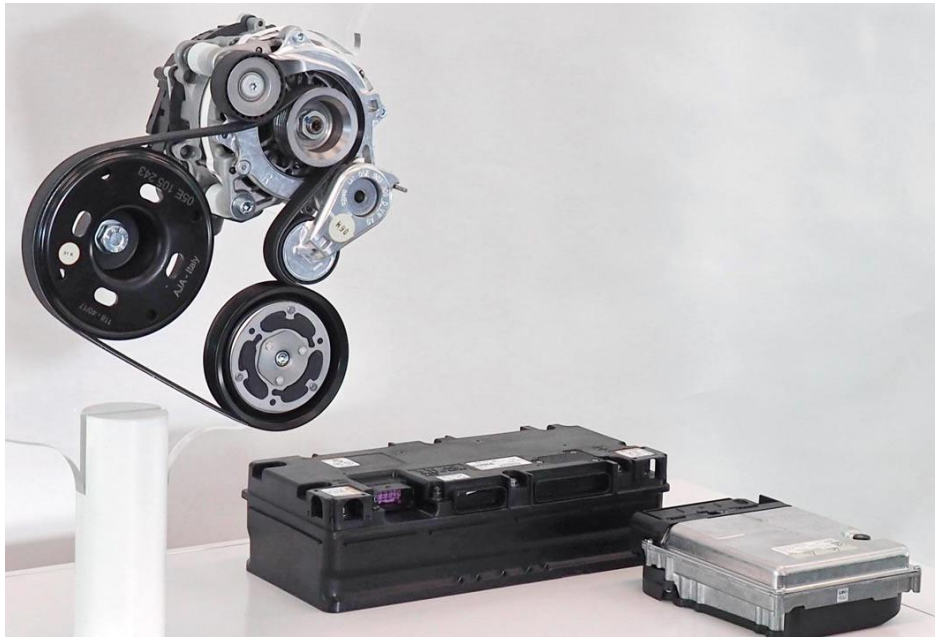


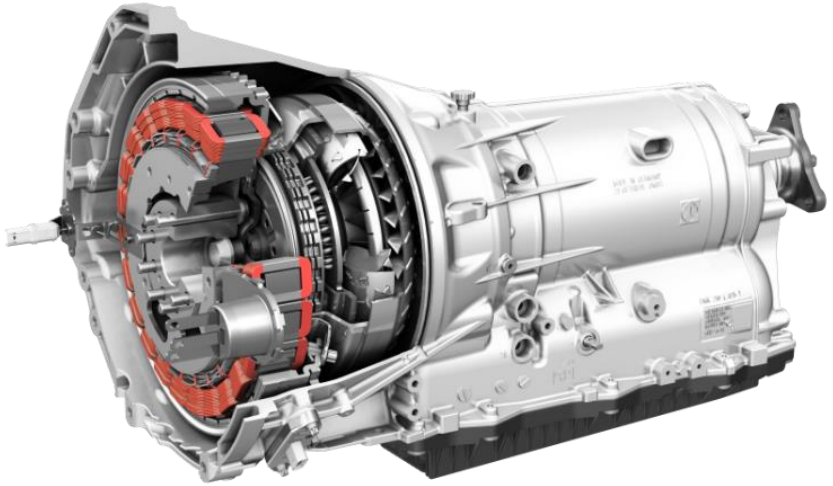
- There will be separate battery (small Li-ion battery pack) for storing electrical energy [1 - 5kW].
- Amount and duration of assistance depends on the size of motor and capacity of battery

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Operation of Mild Hybrids

- Small engines uses belt driven ISG and big engines uses large ISG near flywheel.
- This large ISG will be placed inside the bell housing of transmission with a clutch.

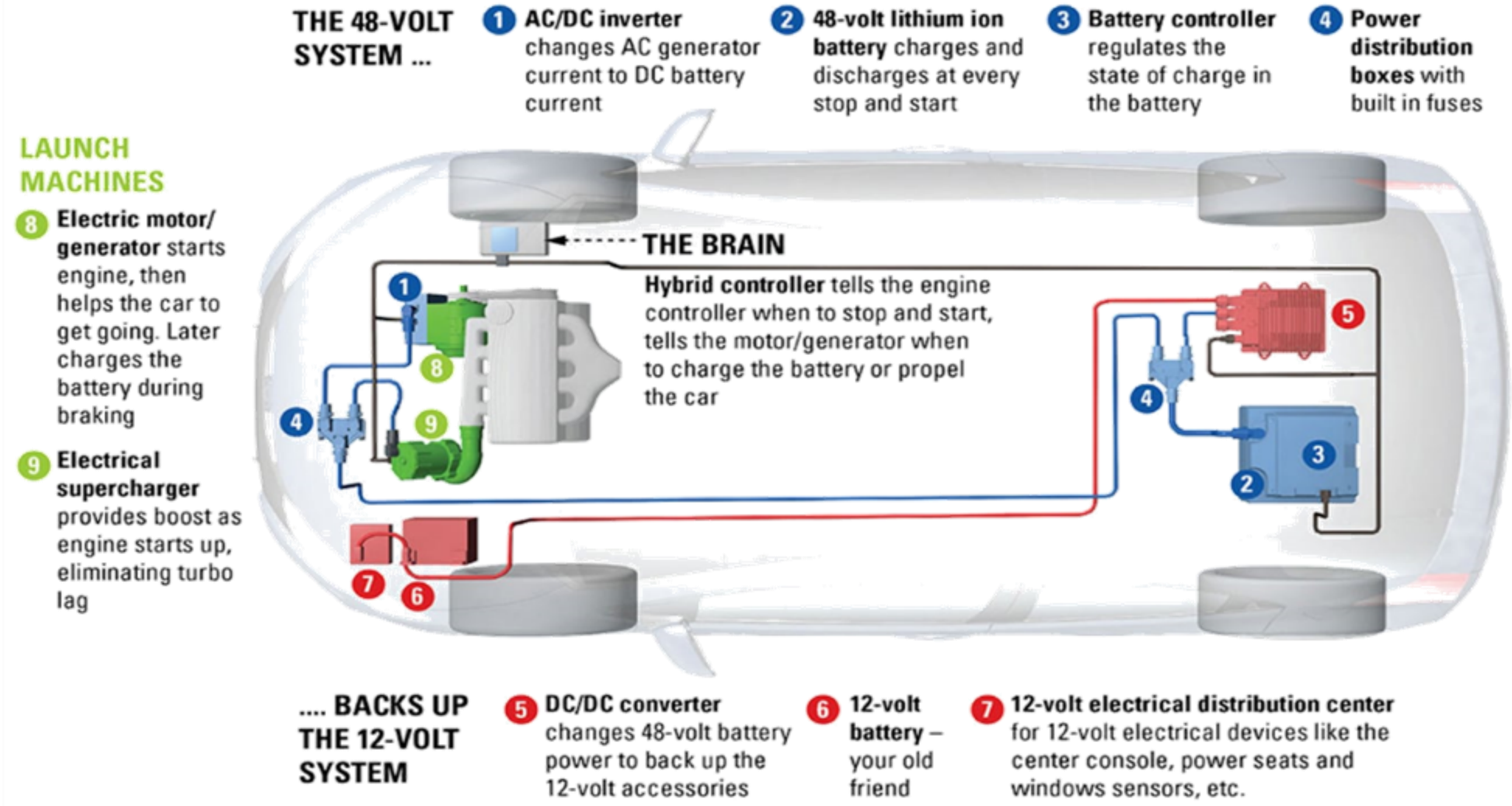




- It also act as generator and charge both batteries and had regenerative braking.
- It is generally a 48V system and also do the job of micro hybrids in stop & go traffic

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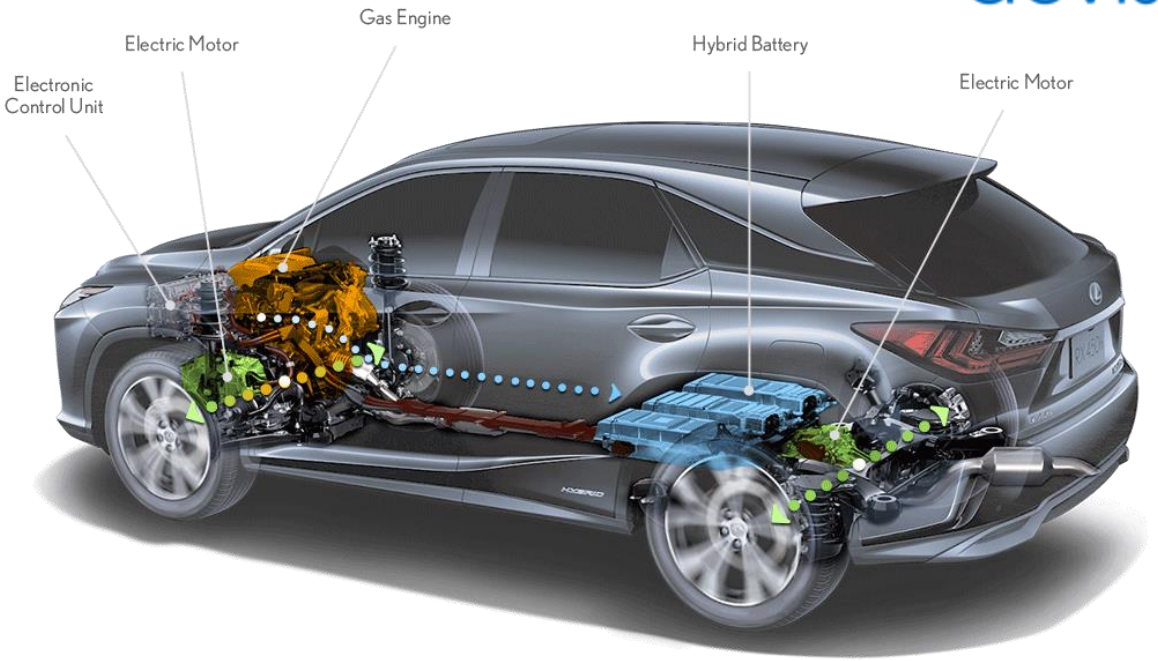
Mild Hybrid standard layout

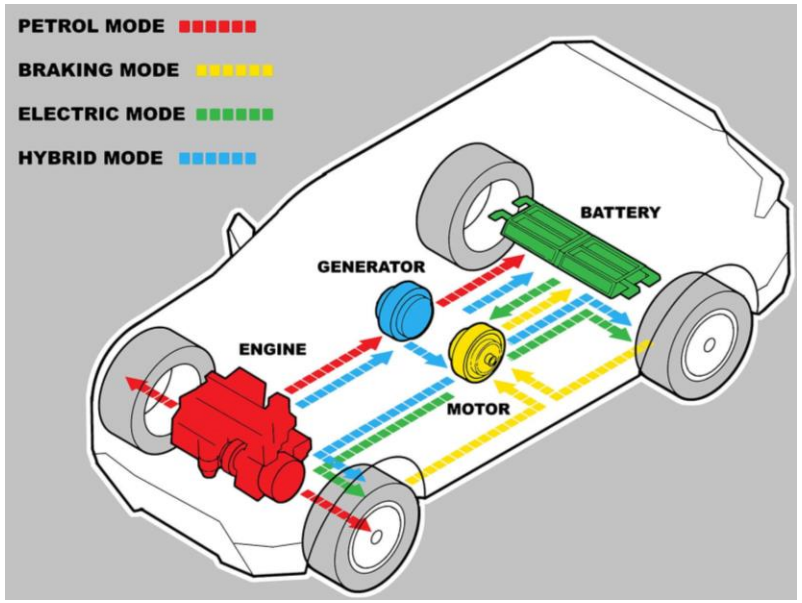


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Full / Strong Hybrids

- This is the vehicle with pure electric driving capacity and flexibility.
- The electric motor is powerful and battery pack is big enough for driving the vehicle



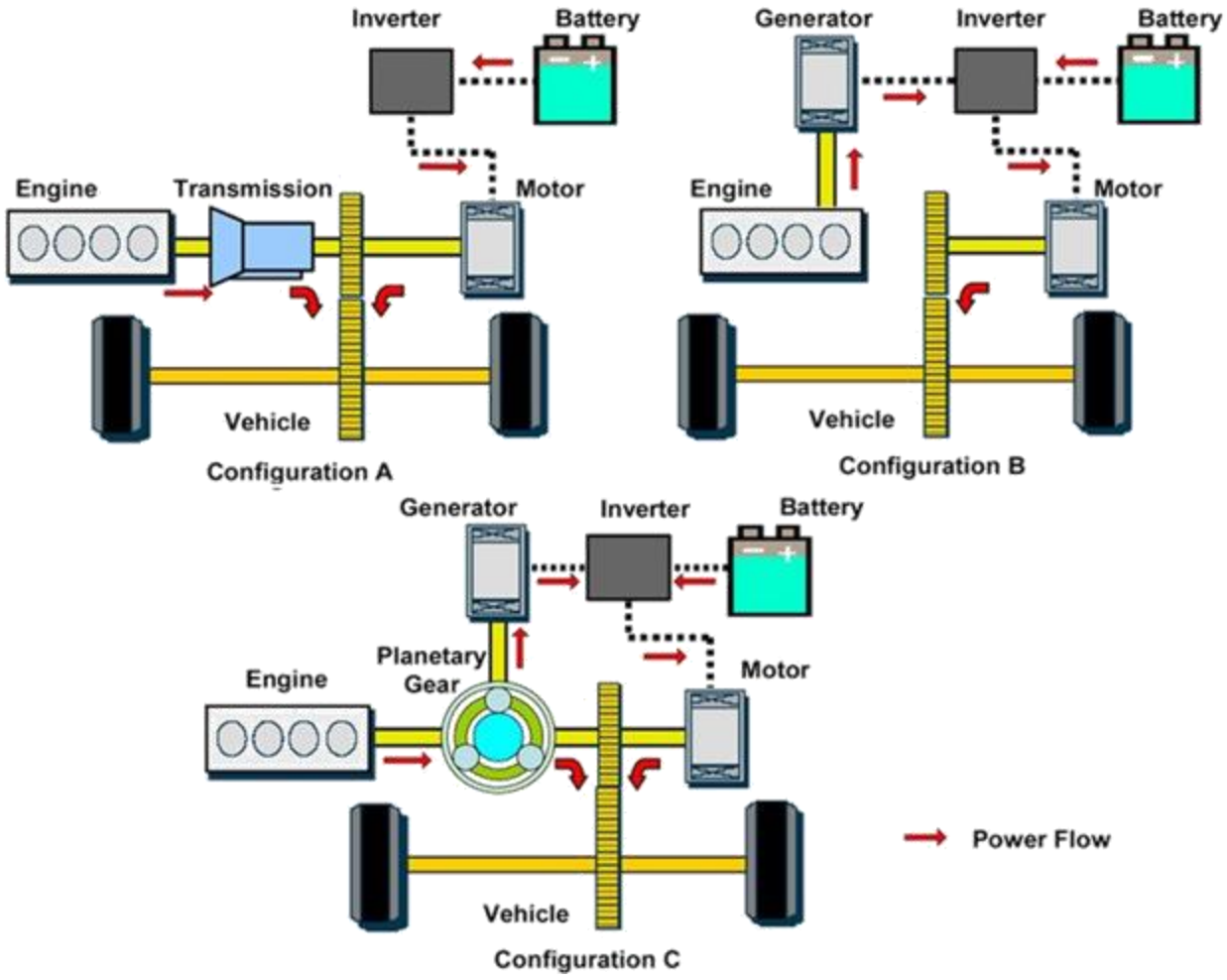


- The engine is now less powerful as the motor can assist it whenever it needed.
- The battery pack is of high voltage and can be hazardous if proper PPE is not used

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Types of strong hybrids

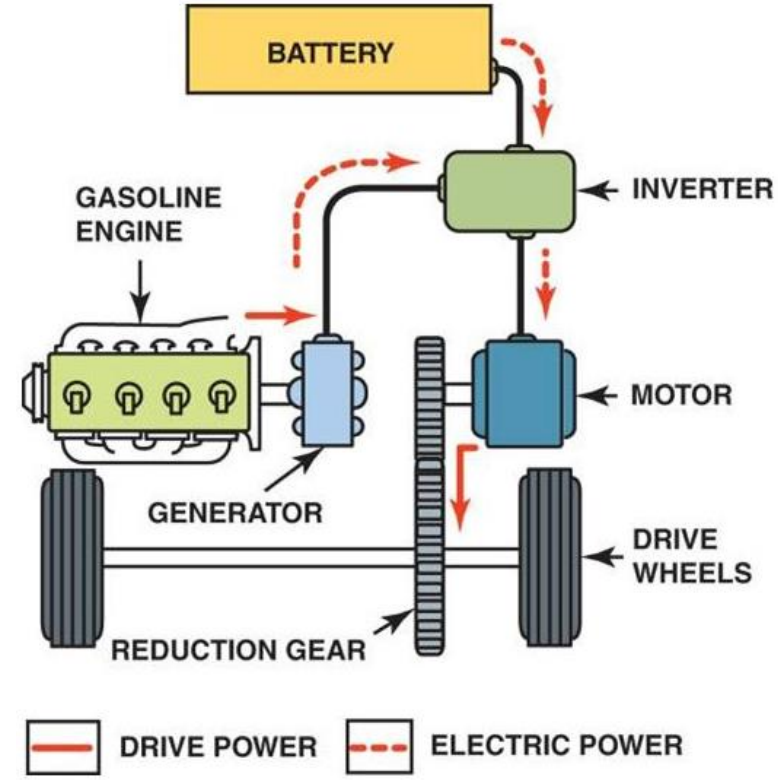
- Depends on drivetrain arrangements there are three configurations.
- Parallel hybrid, where motor drives at low speeds and engine at high speeds (Conf. A)
- Series hybrids, where electric motor drives the wheels & engine recharge battery pack.
- Series-Parallel is a combination of both, and had more flexibility than others (Conf. C)

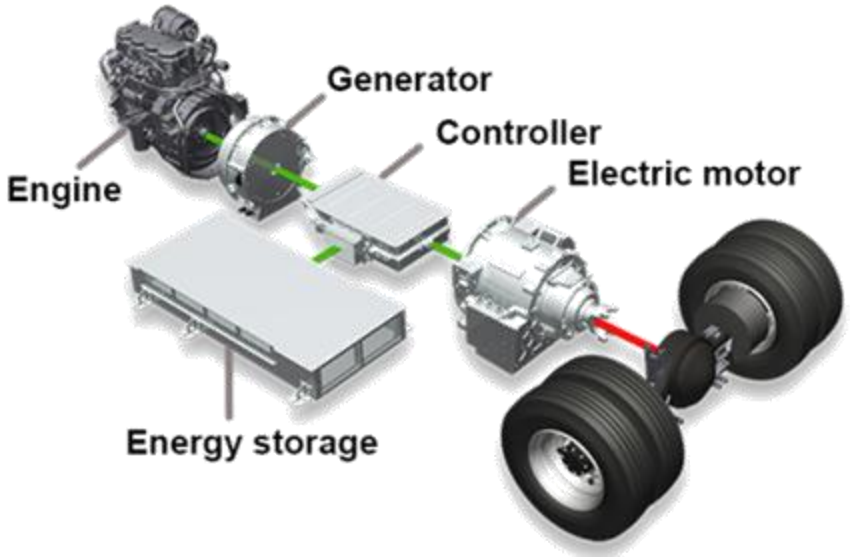


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Series Hybrids – working

- In this IC engine is the primary source of power but it is not driving the wheels.
- The electric motor is responsible for driving the wheels, thus no gearbox needed



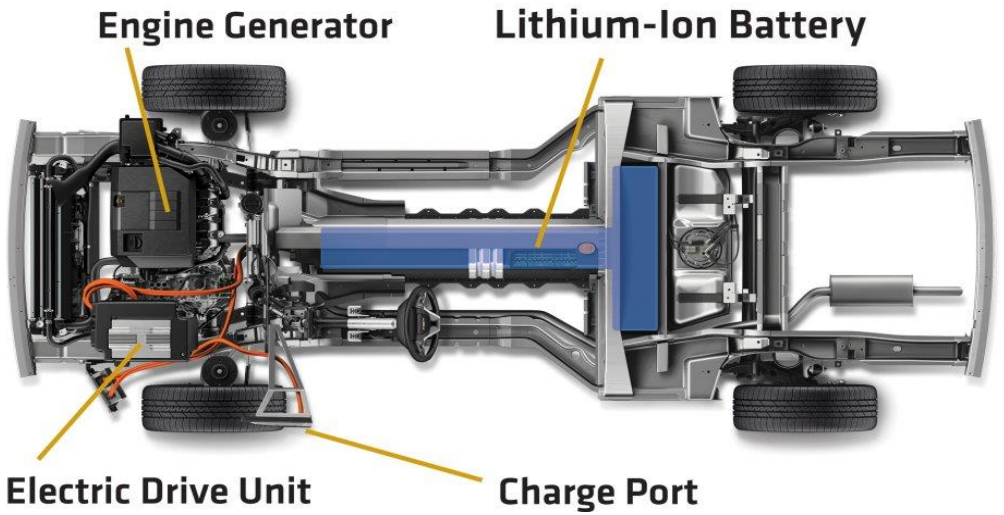


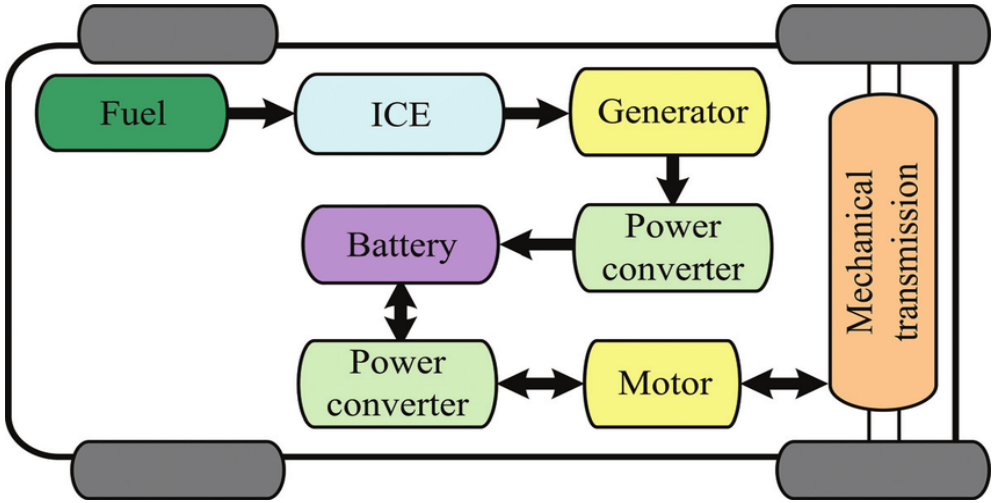
- Engine drives an alternator big enough to recharge the battery pack through invertor.
- The motor takes electricity from this battery pack and convert it into mechanical energy

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Series Hybrid - Efficiency

- IC engine is almost running on its peak efficiency, so low fuel consumption.
- Electric motor provides good torque and battery is always charged via engine





- Total energy efficiency is low, due to multiple energy conversions.
- If the motor/battery is damaged, there is no power available at the wheels for driving

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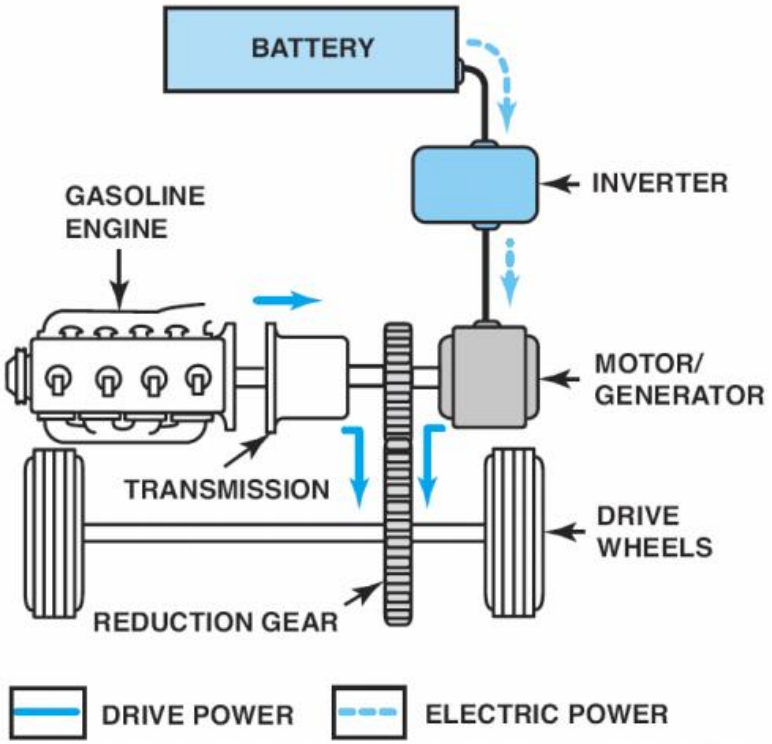
TATA Starbus Hybrid – Series Plugin

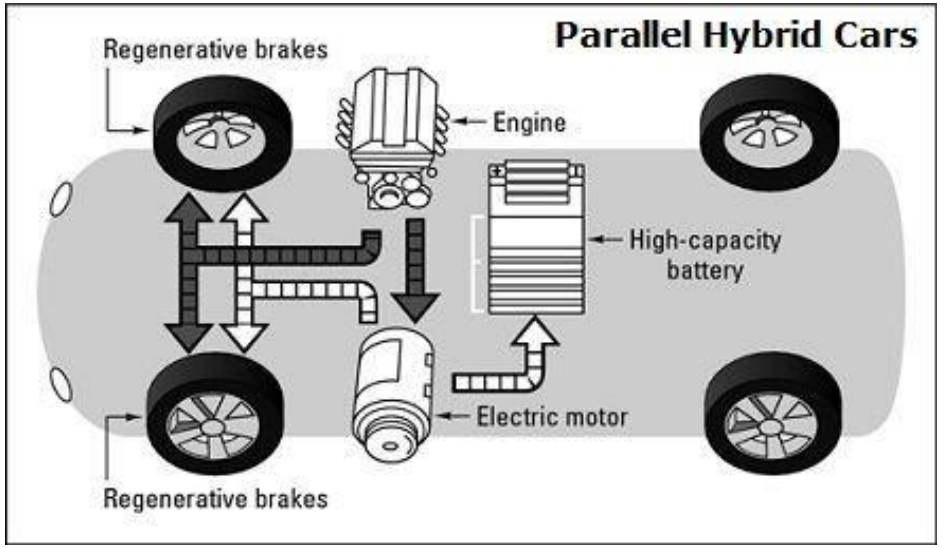


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Parallel Hybrids

- It has both engine and motor independently connected to the drive wheels.
- Electric motor drives vehicle at low speeds and engine take over at high speeds



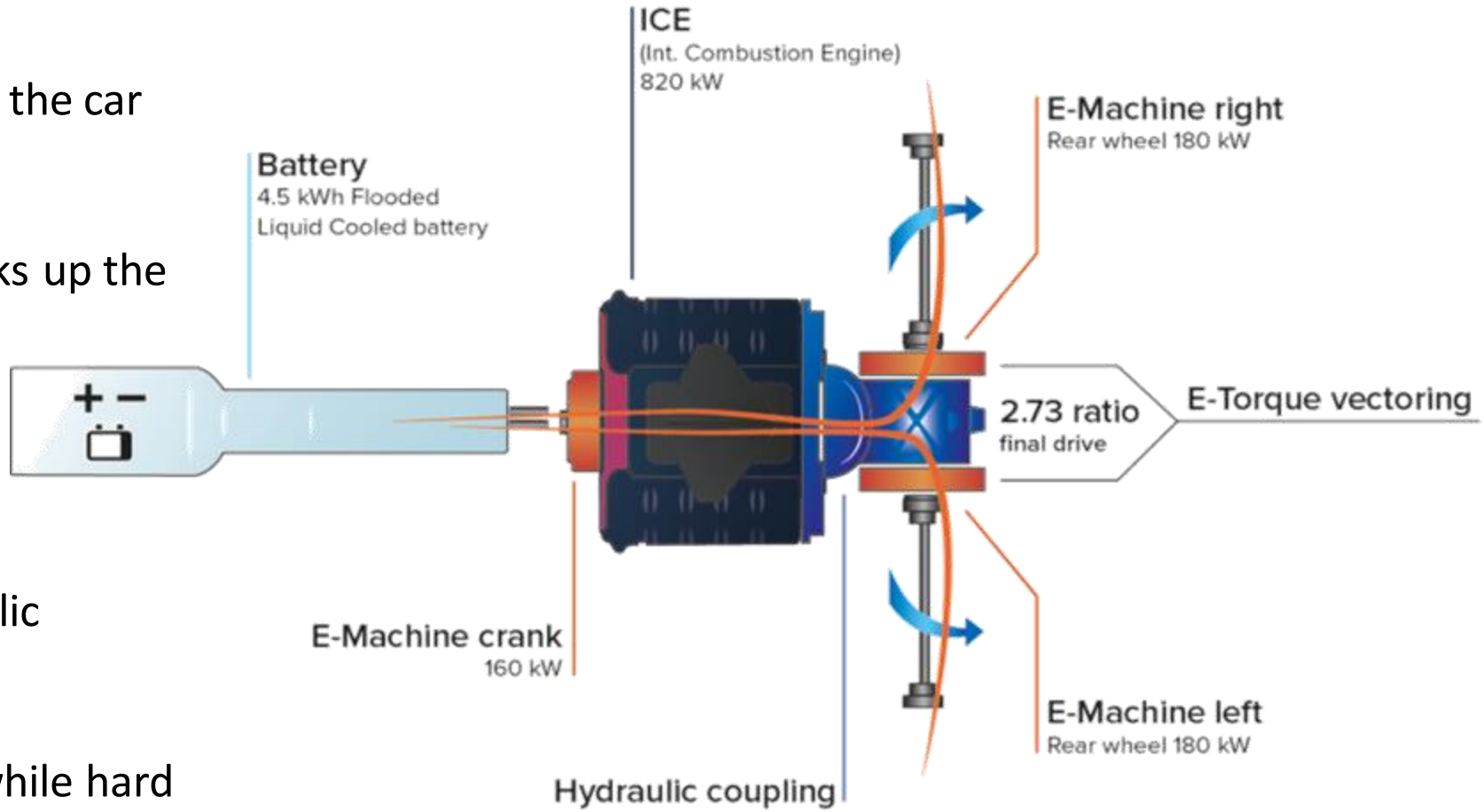


- Both can be placed in the front/rear wheels together or separate, increasing flexibility.
- Electric motor can assist the engine when high torque is required by driver

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Koenigsegg Regera - example

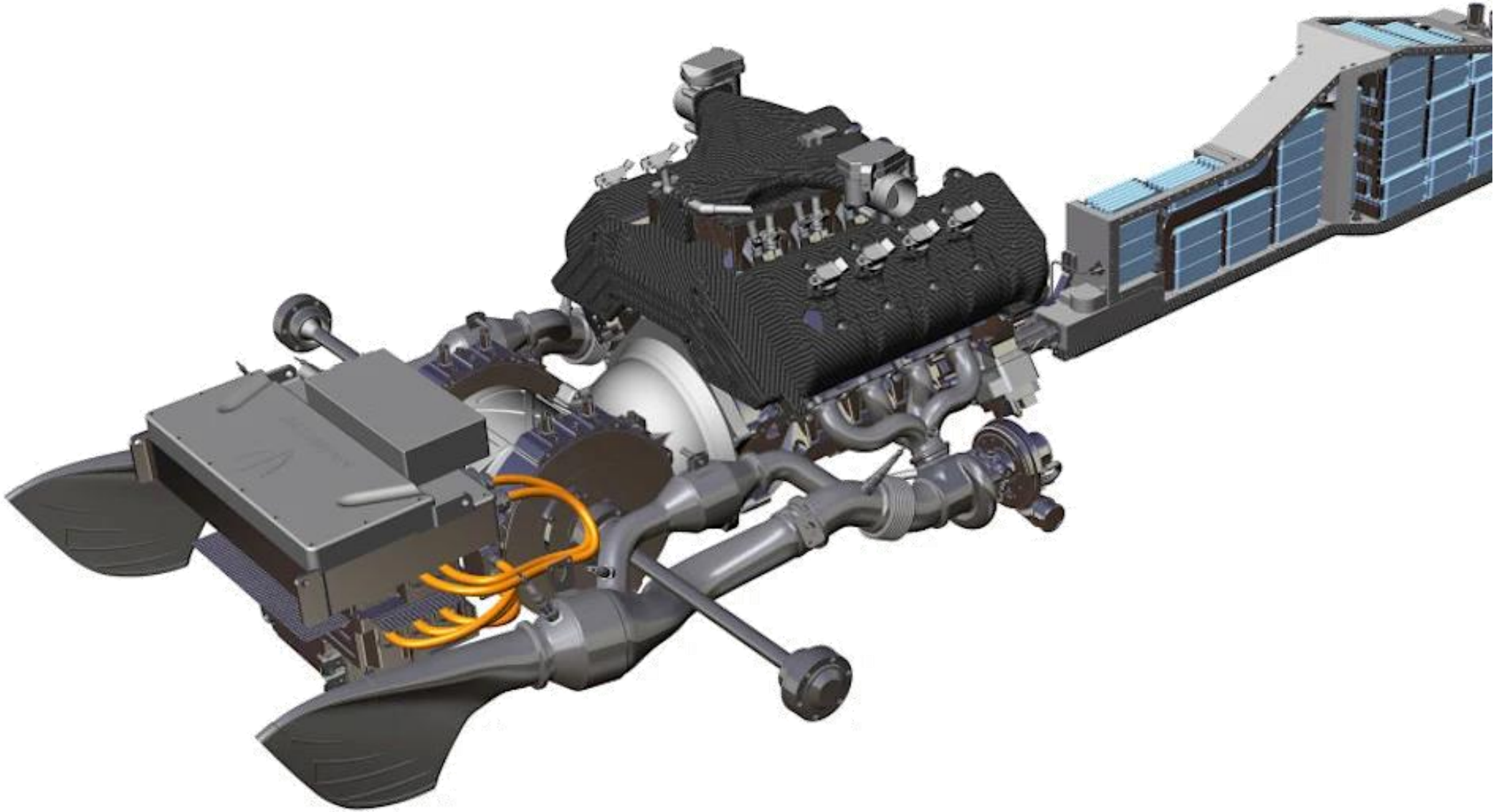
- There is no transmissions but a single gear reduction.
- The 180kW (234HP) motors drive the car up to 30km/hr.
- Then 160kW (208HP) motor cranks up the engine (1066HP).



- Engine drives through the hydraulic coupling
- Electric motors help the engine while hard acceleration

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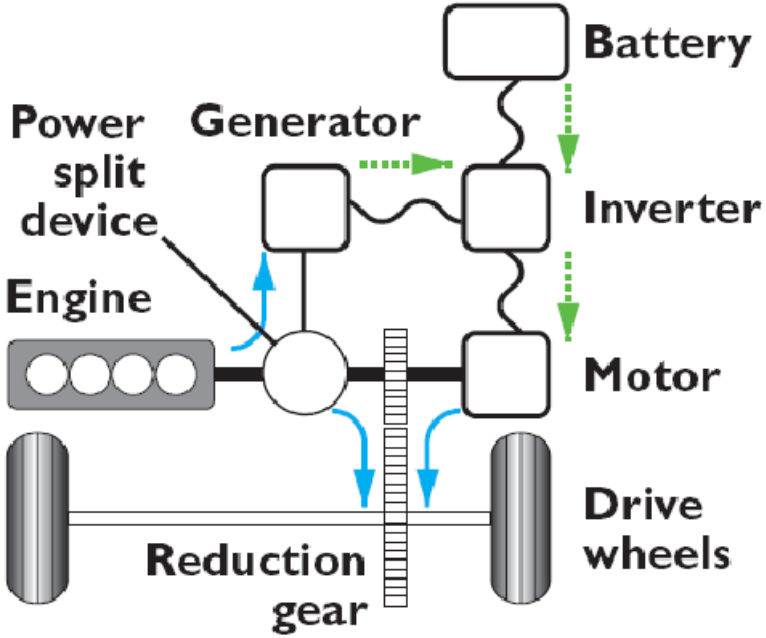
Koengisegg Regera – Powertrain

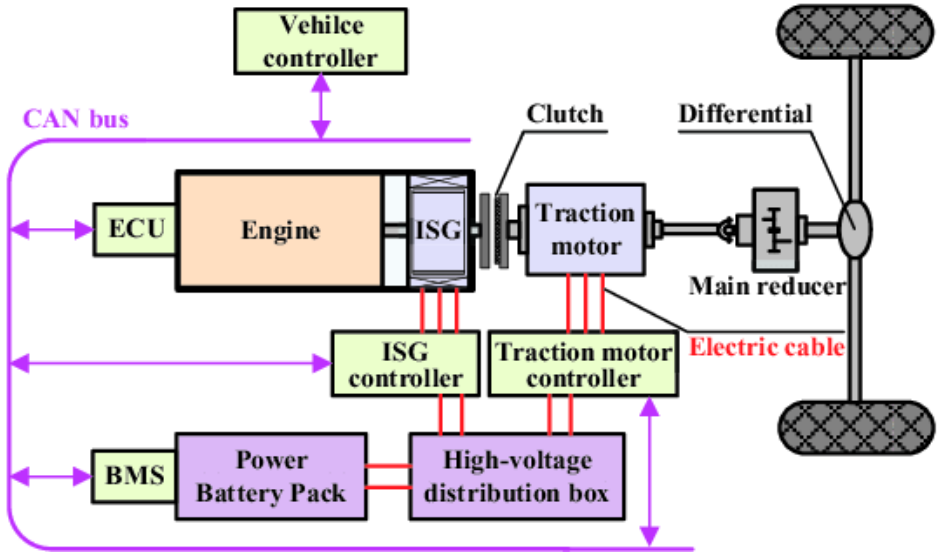


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Series Parallel Hybrids

- It work in series or parallel hybrid configurations so more efficient.
- Power Split device engages or disengages engine/motor/both with drive wheels

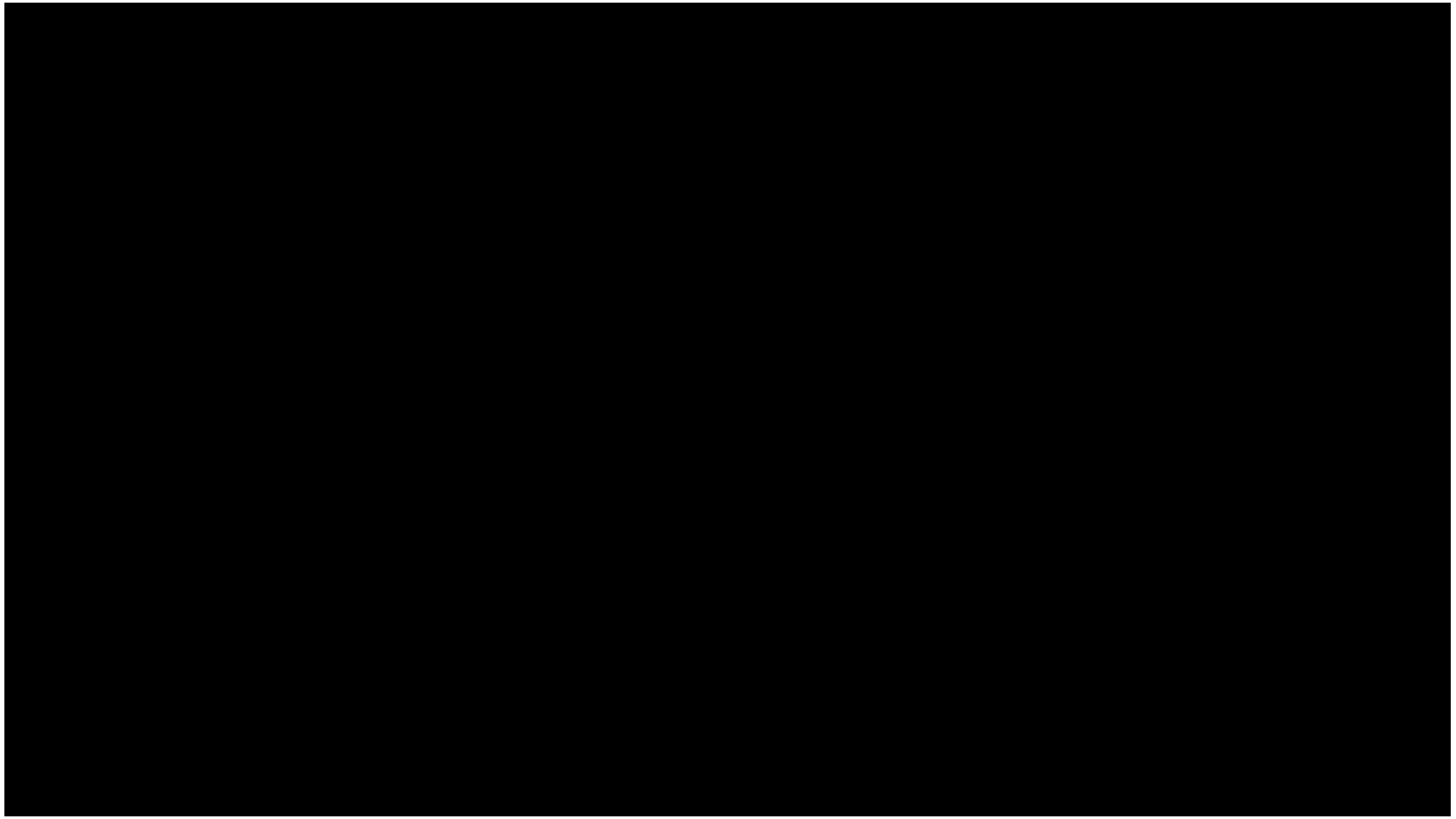




- The engine can charge the battery pack and assist the motor in driving.
- Mostly, motor will be driving the wheels and engine may not need a gearbox for driving

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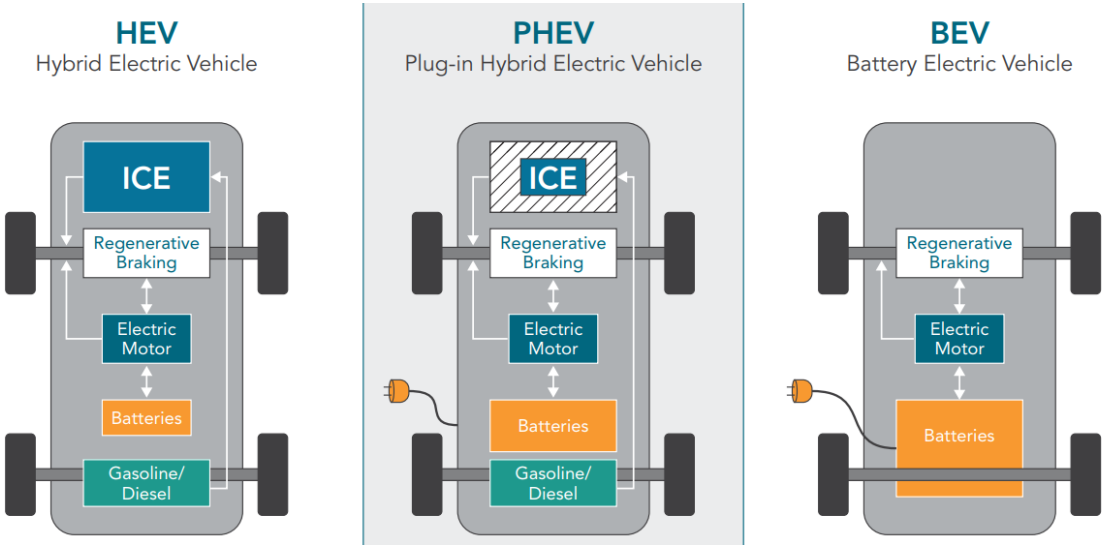
Toyota Hybrid System



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Plug-in Hybrid

- In a conventional hybrid, engine is the only mode of charging the battery pack.
- No provision for charging via external source like house socket/charging station.



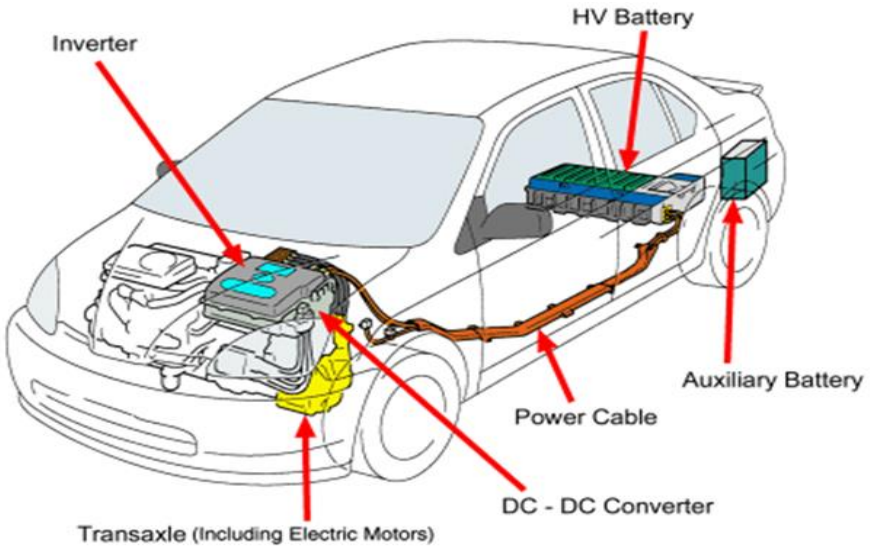


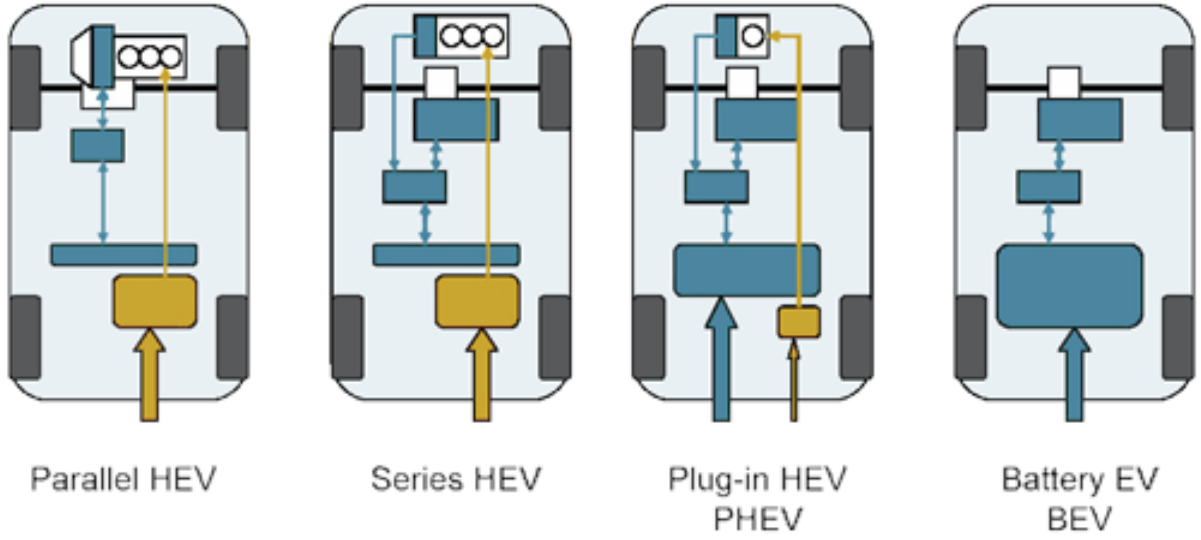
- A plug-in hybrid can utilise an external source to charge the battery pack.
- Thus, it does not need to switch on the engine in small city driving

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Hybrids – Feature list

Classification/ function	Start/stop	Regeneration	Electrical assistance	Electric-only driving	Charging from a power socket
Start/stop system	√	√			
Mild hybrid	√	√	√		
Strong hybrid	√	√	√	√	
Plug-in hybrid	√	√	√	√	√





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